Case study 5:-

Bank Interest Calculator :-A bank wants to estimate the Simple Interest earned by customers based On Principal Amount, Rate of Interest, and Time Period.

Algorithm – Bank Interest Calculator

1. Start.
2. Declare variables: principal, rate, time, interest.
3. Input the Principal Amount, Rate of Interest, and Time.
4. Calculate Simple Interest using the formula:
    - interest = (principal × rate × time) / 100
5. Display the Simple Interest.
6. Stop.

Pseudocode:-

```
BEGIN

DECLARE principal, rate, time, interest AS FLOAT

INPUT principal
INPUT rate
INPUT time

interest = (principal * rate * time) / 100

PRINT "Simple Interest = ", interest

END
```

Test cases:- input/output combination.

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_5.c -o CASE_study_5 ; if ($?) { .\CASE_study_5 }
enter the principle 50000
enter the rate 7
enter the time 3
simple_interest=10500.000000
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_5.c -o CASE_study_5 ; if ($?) { .\CASE_study_5 }
enter the principle 70000
enter the rate 8
enter the time 4
simple_interest=22400.000000
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_5.c -o CASE_study_5 ; if ($?) { .\CASE_study_5 }
enter the principle 30000
enter the rate 5
enter the time 2
simple_interest=3000.000000
PS C:\Users\khush\Desktop\PROJECT 1>
```